



AI Research Group

CENTRE FOR ADVANCED STUDIES

Dr. A.P.J. Abdul Kalam Technical University
Lucknow



Research Seminar Series

Seminar -5

Cost Function Optimization Strategies in Deep Neural Network

DATE & TIME:
3rd February 2021,
4:00 PM.

PLATFORM & WEB-LINK
Google Meet :
<https://meet.google.com/ynq-yqdn-zkx>



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ABSTRACT

Nowadays, deep learning which is a contemporary and interesting field of machine learning made a significant revolution with appreciable performance in a variety of applications such as medical image classification, natural language processing, face recognition etc. It is considered to be the best choice for discovering complex architecture in high-dimensional data by using the backpropagation algorithm. The optimizer is one of the core techniques behind the accomplishment of advanced deep neural networks. An efficient way of deep network optimization is to make adaptive step sizes for each parameter. Recently, several efforts have been made to improve gradient descent methods such as AdaGrad, AdaDelta, RMSProp and Adam. These methods depend on the square roots of exponential moving averages of squared past gradients. Thus, these methods do not take the advantage of local change in gradients. A new optimizer “diffGrad” has been discussed which is based on the difference between the current and the immediate past gradient. In diffGrad optimization technique, the step size is adjusted for each parameter in such a way that it should have a larger step size for faster gradient shifting parameters and lower step size for lower gradient shifting parameters.